



Magula Plan

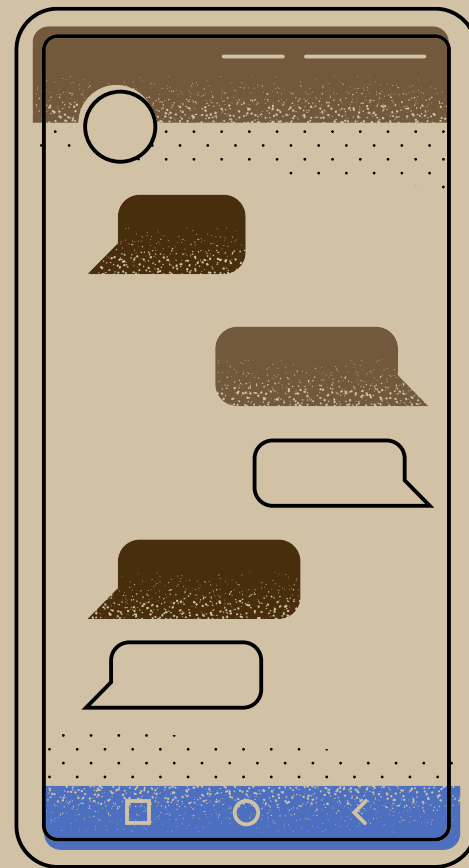
A Centralized Digital Wedding Planning
Platform for Sri Lanka

Start Presentation

Group 04: Maheen, Amanda, Dileepa, Sammani, Ruchira



PROBLEM & RESEARCH



Culturally vast, digitally scattered

- Vendor search spread across Facebook, Instagram, WhatsApp – no compare-by-price
- Budgets on paper: no deposit/balance tracking, no overspend alerts
- Guest lists & RSVPs run one phone call at a time – never a live headcount
- No local way to send a digital invite + RSVP link over WhatsApp
- Global tools (The Knot, Zola) aren't built for multi – day Sri Lankan ceremonies

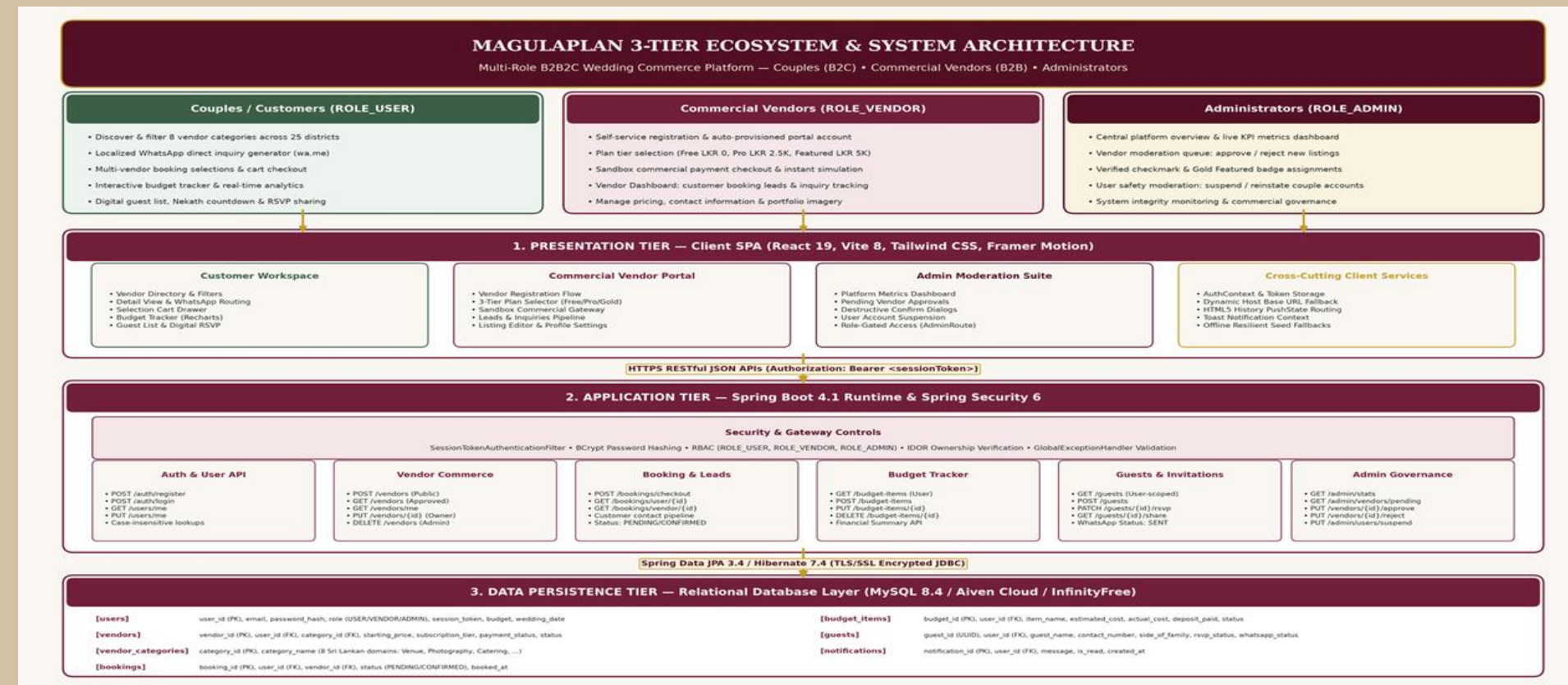
SOLUTION & TARGET MARKET



One connected workflow

- Discover vendors · track budget · manage guests · send invites
- MVP: Vendor Directory, Budget Tracker, Guest & RSVP, Web Share invitations
- Plus couple dashboard, admin moderation, multi-vendor cart, commercial vendor portal
- 8 vendor categories · 13 verified Sri Lankan vendors seeded
- B2B2C marketplace: For engaged couples and local wedding vendors – Poruwa & Nekath aware

TECH STACK & ARCHITECTURE



A 3-tier layered architecture

- Presentation → Application → Data
- React 19 · Vite · Tailwind 3.4 · Framer Motion · Recharts · Web Share API
- Spring Boot 4.1 REST · Spring Security 6 RBAC · IDOR Protection · BCrypt · session-token filter
- MySQL 8.4 · JPA/Hibernate · 7 normalized tables · idempotent Startup seeding





HIGH LEVEL ARCHITECT DIAGRAM

MAGULAPLAN 3-TIER ECOSYSTEM & SYSTEM ARCHITECTURE

Multi-Role B2B2C Wedding Commerce Platform — Couples (B2C) • Commercial Vendors (B2B) • Administrators

Couples / Customers (ROLE_USER)

- Discover & filter 8 vendor categories across 25 districts
- Localized WhatsApp direct inquiry generator (wa.me)
- Multi-vendor booking selections & cart checkout
- Interactive budget tracker & real-time analytics
- Digital guest list, Nekath countdown & RSVP sharing

Commercial Vendors (ROLE_VENDOR)

- Self-service registration & auto-provisioned portal account
- Plan tier selection (Free LKR 0, Pro LKR 2.5K, Featured LKR 5K)
- Sandbox commercial payment checkout & instant simulation
- Vendor Dashboard: customer booking leads & inquiry tracking
- Manage pricing, contact information & portfolio imagery

Administrators (ROLE_ADMIN)

- Central platform overview & live KPI metrics dashboard
- Vendor moderation queue: approve / reject new listings
- Verified checkmark & Gold Featured badge assignments
- User safety moderation: suspend / reinstate couple accounts
- System integrity monitoring & commercial governance

1. PRESENTATION TIER — Client SPA (React 19, Vite 8, Tailwind CSS, Framer Motion)

Customer Workspace

- Vendor Directory & Filters
- Detail View & WhatsApp Routing
- Selection Cart Drawer
- Budget Tracker (Recharts)
- Guest List & Digital RSVP

Commercial Vendor Portal

- Vendor Registration Flow
- 3-Tier Plan Selector (Free/Pro/Gold)
- Sandbox Commercial Gateway
- Leads & Inquiries Pipeline
- Listing Editor & Profile Settings

Admin Moderation Suite

- Platform Metrics Dashboard
- Pending Vendor Approvals
- Destructive Confirm Dialogs
- User Account Suspension
- Role-Gated Access (AdminRoute)

Cross-Cutting Client Services

- AuthContext & Token Storage
- Dynamic Host Base URL Fallback
- HTML5 History PushState Routing
- Toast Notification Context
- Offline Resilient Seed Fallbacks

HTTPS RESTful JSON APIs (Authorization: Bearer <sessionToken>)

2. APPLICATION TIER — Spring Boot 4.1 Runtime & Spring Security 6

Security & Gateway Controls

SessionTokenAuthenticationFilter • BCrypt Password Hashing • RBAC (ROLE_USER, ROLE_VENDOR, ROLE_ADMIN) • IDOR Ownership Verification • GlobalExceptionHandler Validation

Auth & User API

- POST /auth/register
- POST /auth/login
- GET /users/me
- PUT /users/me
- Case-insensitive lookups

Vendor Commerce

- POST /vendors (Public)
- GET /vendors (Approved)
- GET /vendors/me
- PUT /vendors/{id} (Owner)
- DELETE /vendors (Admin)

Booking & Leads

- POST /bookings/checkout
- GET /bookings/user/{id}
- GET /bookings/vendor/{id}
- Customer contact pipeline
- Status: PENDING/CONFIRMED

Budget Tracker

- GET /budget-items (User)
- POST /budget-items
- PUT /budget-items/{id}
- DELETE /budget-items/{id}
- Financial Summary API

Guests & Invitations

- GET /guests (User-scoped)
- POST /guests
- PATCH /guests/{id}/rsvp
- GET /guests/{id}/share
- WhatsApp Status: SENT

Admin Governance

- GET /admin/stats
- GET /admin/vendors/pending
- PUT /vendors/{id}/approve
- PUT /vendors/{id}/reject
- PUT /admin/users/suspend

Spring Data JPA 3.4 / Hibernate 7.4 (TLS/SSL Encrypted JDBC)

3. DATA PERSISTENCE TIER — Relational Database Layer (MySQL 8.4 / Aiven Cloud / InfinityFree)

[users]	user_id (PK), email, password_hash, role (USER/VENDOR/ADMIN), session_token, budget, wedding_date
[vendors]	vendor_id (PK), user_id (FK), category_id (FK), starting_price, subscription_tier, payment_status, status
[vendor_categories]	category_id (PK), category_name (8 Sri Lankan domains: Venue, Photography, Catering, ...)
[bookings]	booking_id (PK), user_id (FK), vendor_id (FK), status (PENDING/CONFIRMED), booked_at

[budget_items]	budget_id (PK), user_id (FK), item_name, estimated_cost, actual_cost, deposit_paid, status
[guests]	guest_id (UUID), user_id (FK), guest_name, contact_number, side_of_family, rsvp_status, whatsapp_status
[notifications]	notification_id (PK), user_id (FK), message, is_read, created_at

PROJECT PLAN & RISK MANAGEMENT

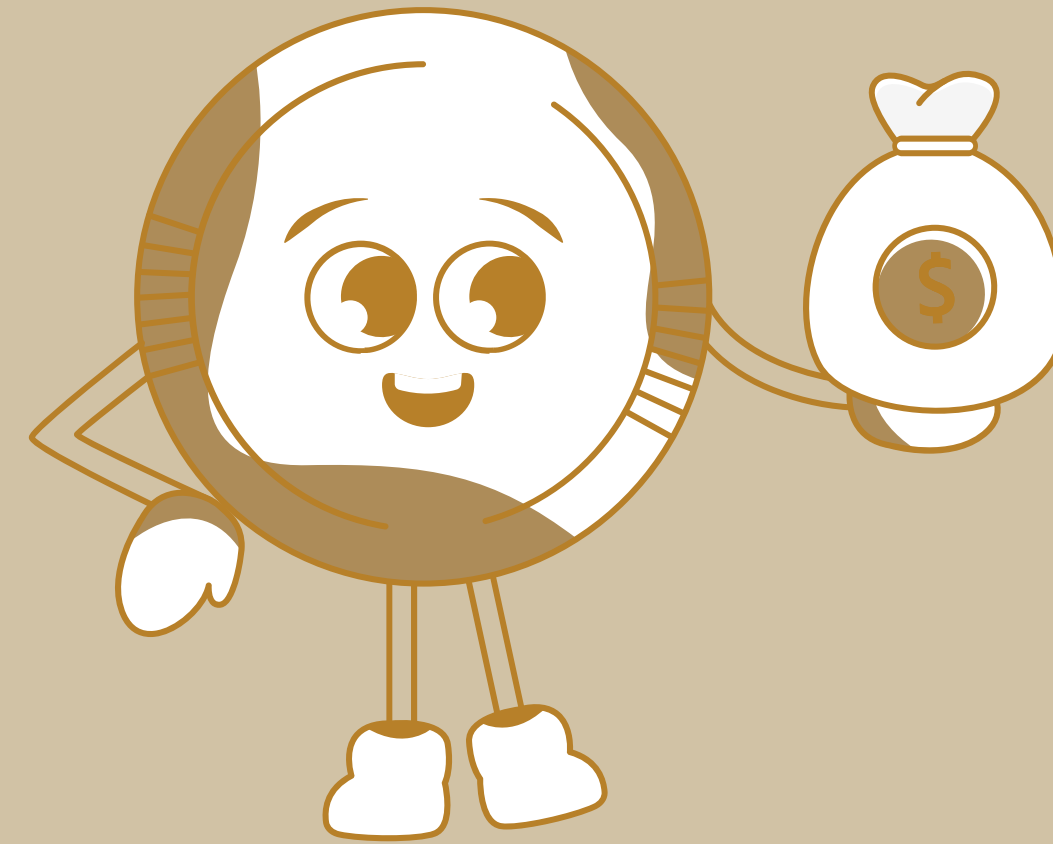


Agile — 3 × 2-week sprints over 6 weeks

- S1 design & setup · S2 core features · S3 integrate, test, deploy
- 31 / 31 Jira tickets closed · Kanban + PR reviews + async standups
- Risk: Meta WhatsApp API cost → pivoted to Web Share API
- Risk: free-tier DB sleep → retry pooling + offline fallback
- Risk: plaintext passwords → BCrypt + RBAC
- Risk: live payment gateway scope creep → solved via simulated payment sandbox



BUSINESS MODEL & BUDGET



LKR 861,500 planned → LKR 720,000 actual

- 16% saved – spend is engineering hours only; all cloud & tools on free tiers
- Revenue 1: freemium vendor plans –Free LKR 0 / Pro LKR 2,500 / Featured LKR 5,000 per month
- Revenue 2: 2.5–5% commission on confirmed bookings
- Revenue 3: category sponsorships & featured banners
- Monetization model validated via interactive simulated payment sandbox





MAHEEN SAYURU

CIT-24-02-0189

21%



Project Manager · Full-Stack / DevOps · 9 Jira tickets

- Contribution assessed across the full SDLC – tickets, tests, docs – not raw commits
- Authored the PM Plan, WBS & Jira sprint governance
- Built the GitHub monorepo + InfinityFree & Render deploy pipelines
- Shipped budget-summary & RSVP PATCH APIs + Web Share invitations



K.A.R.D. SAMMANI

CIT-24-02-0058

19%

Database Engineer · Business Analyst · 4 Jira tickets

- Designed the 3NF MySQL 8.4 schema + ER diagram
- Wrote idempotent production seed scripts with real vendor data
- Verified the Aiven cloud migration; authored functional & non-functional specs





AMANDA LAKMAL

CIT-24-02-0007

21%

Backend Lead Developer • 7 Jira tickets

- Layered Architecture: Built Spring Boot backend using Controller, Service, and Repository patterns.
- REST APIs: Developed core CRUD endpoints (Users, Vendors, Budget, Guests).
- System Security: Implemented Spring Security, BCrypt hashing, and Session Tokens.
- Error Handling: Integrated Global Exception Handling for consistent API responses.





DILEEPA RANATHUNGA

CIT-24-02-0046

20%



UI/UX Lead · Frontend Developer · 8 Jira tickets

- **Core Frontend:** Built the responsive React 19 SPA architecture and reusable components.
- **Multi-Vendor Cart Drawer:** Developed the cart system for seamless vendor shortlisting and booking.
- **Admin Dashboard:** Created the centralized dashboard for platform monitoring and user/vendor management.
- **Landing & Login UI:** Designed the vintage-themed Landing and Login pages featuring a responsive full-screen hero video.



RUCHIRA NIMNAKA

CIT-24-02-0029

19%

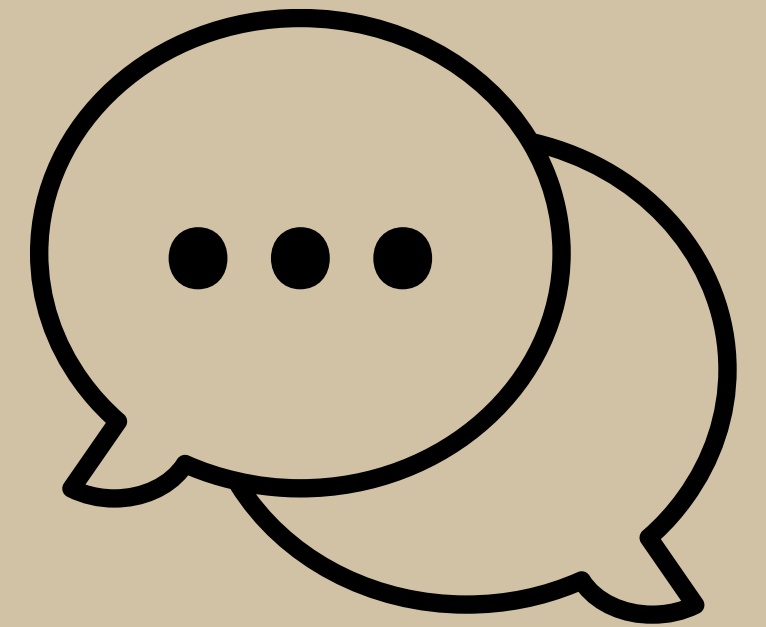
QA Lead · Test Automation Engineer · 3 Jira tickets

- Ran & verified the 142-test JUnit suite (100% pass)
- Authored the 55-scenario System Test Plan + Playwright E2E suite
- Maintained the Defect Register: logged, retested, verified fixes





FUTURE UPGRADES



A three-phase roadmap

Phase 1:

Q4 2026 – Production PayHere/Stripe gateway with LankaQR, live escrow, auto PDF invoicing.

Phase 2:

Q1 2027 – Live vendor availability calendar (Nekath sync), in-app WebSocket chat.

Phase 3:

Q2 2027 – AI Nekath & itinerary generator, 2D drag-drop seating chart, Sinhala/Tamil/English.







Thank You